



PACE AI

Intelligent Campus Information Assistant

A source-grounded RAG system for official campus information

[Student/team names] · Guide: [Guide name]

[Department] · [Academic year]

Problem and proposed solution

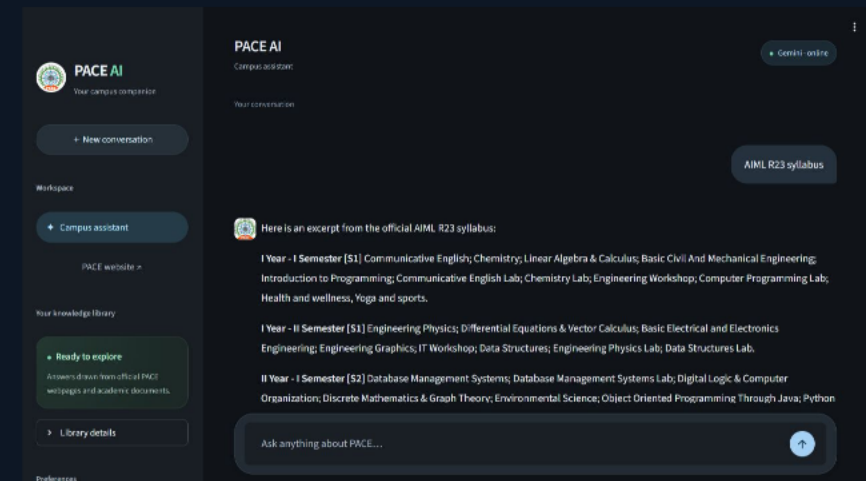
Before PACE AI

Information spread across
webpages and long PDFs

Syllabi and regulations
Admissions and placements
Repeated manual searching

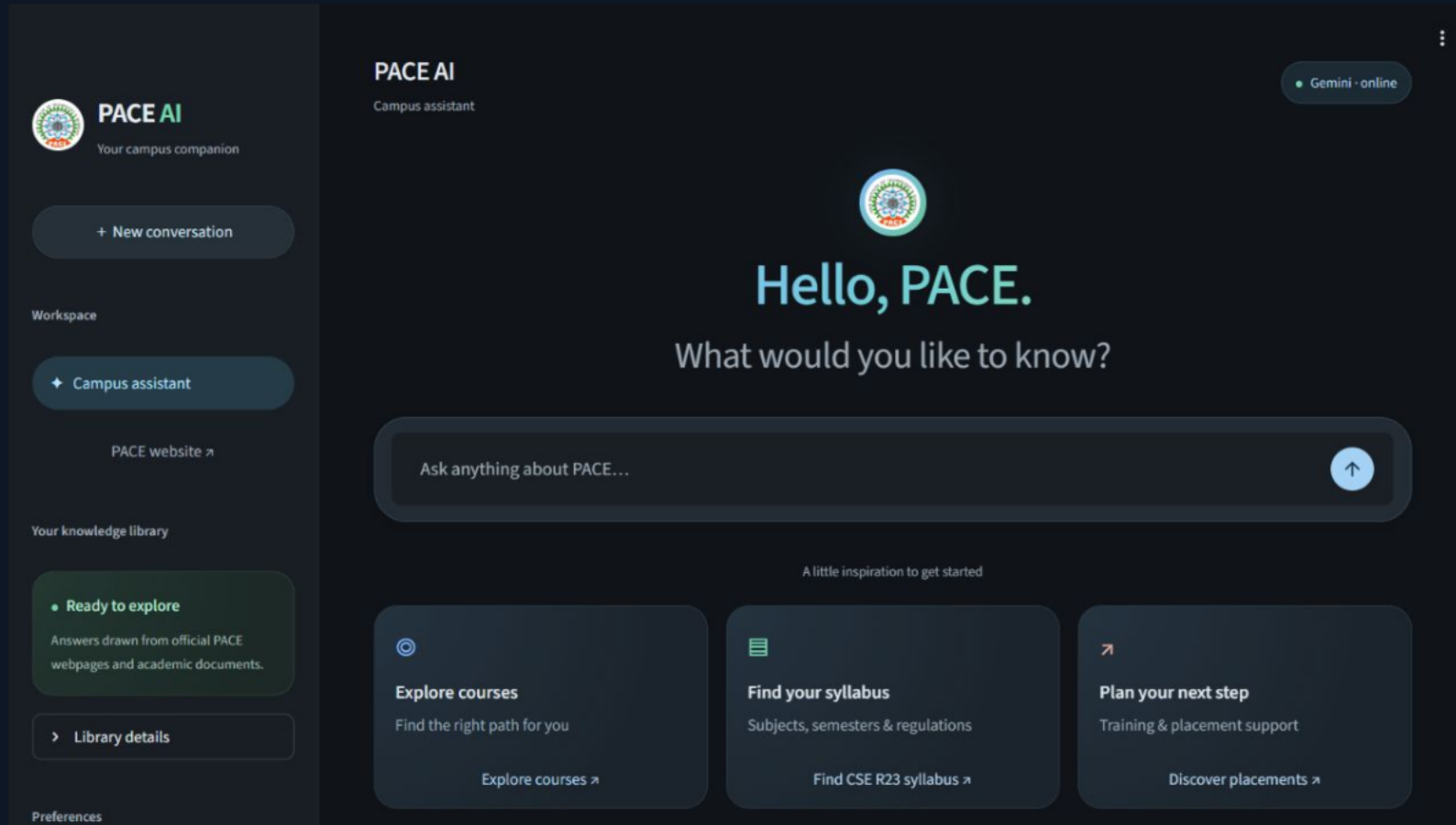
With PACE AI

One question with official evidence



Answer text links back to the exact
webpage or PDF page.

Current interface and features



Grounded answers

Gemini streaming
Evidence-based refusals

Relevant retrieval

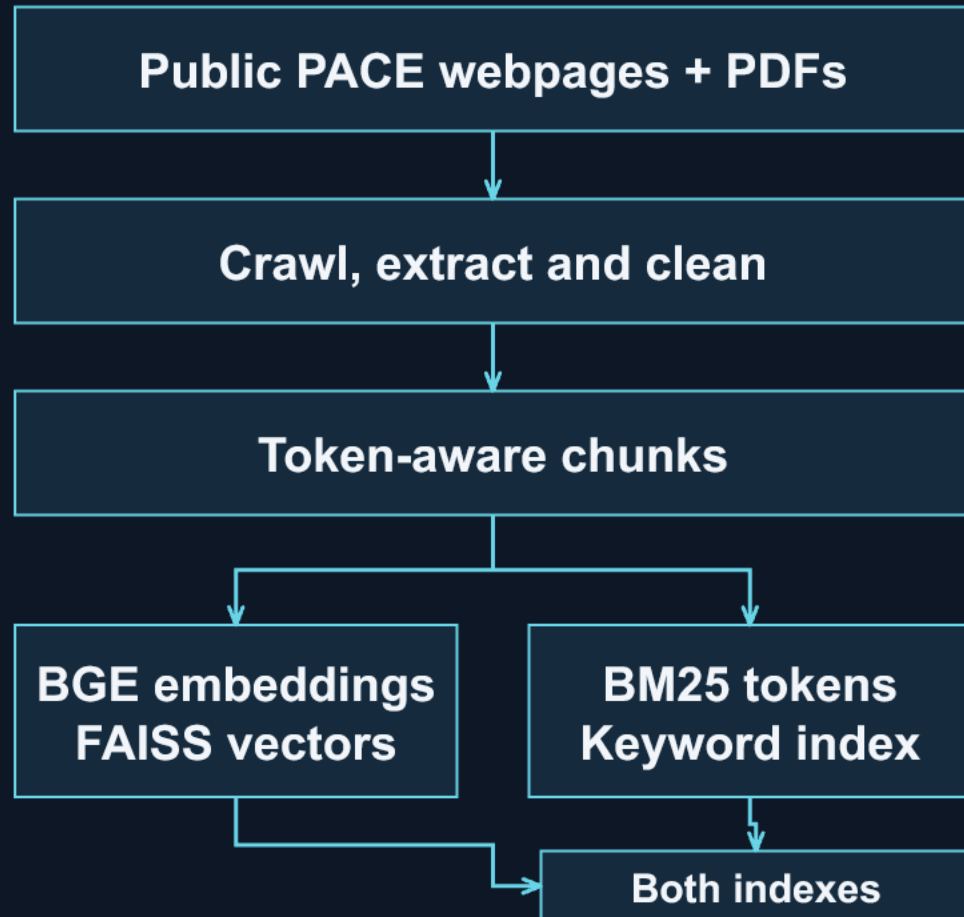
FAISS + BM25
Branch and regulation filters
Token-aware chunks

Traceable results

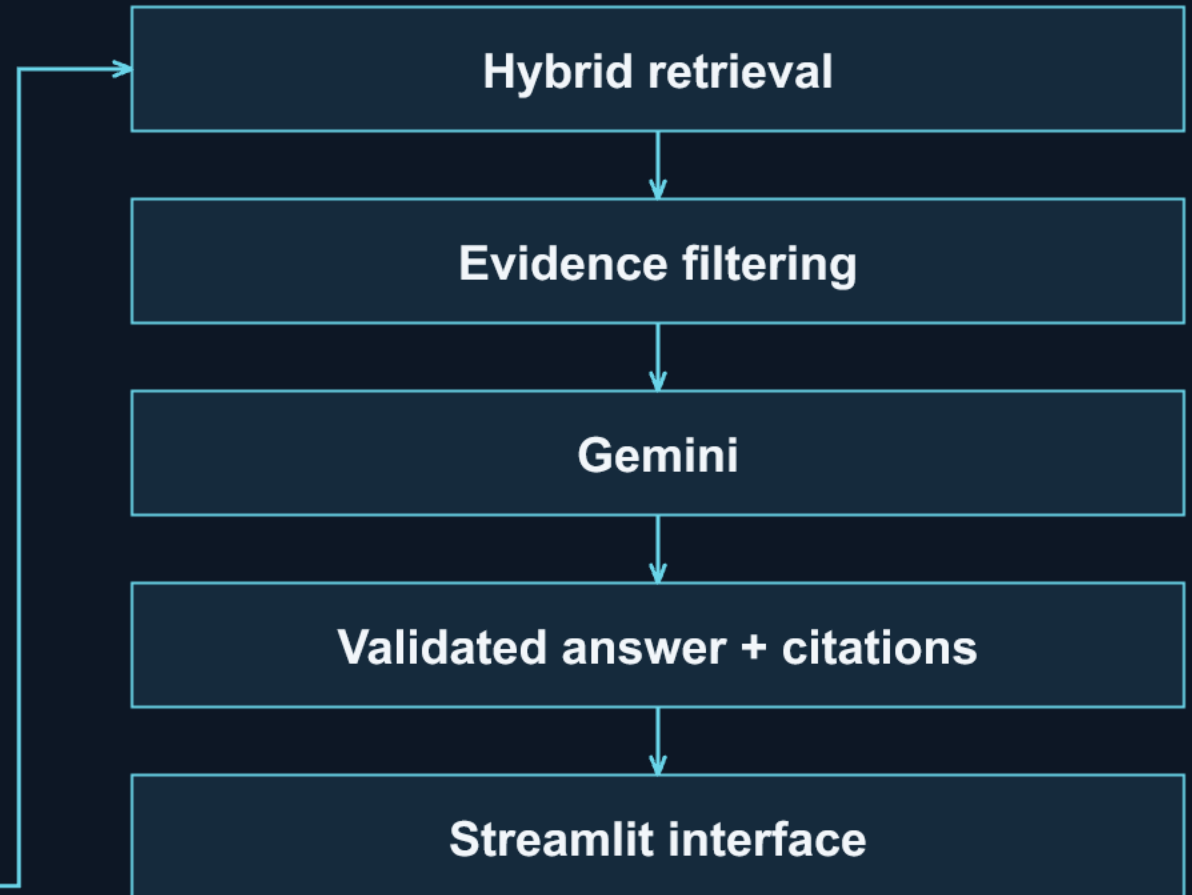
PDF pages and source cards
Retrieval cache
Safe index rollback

System architecture

OFFLINE PREPARATION

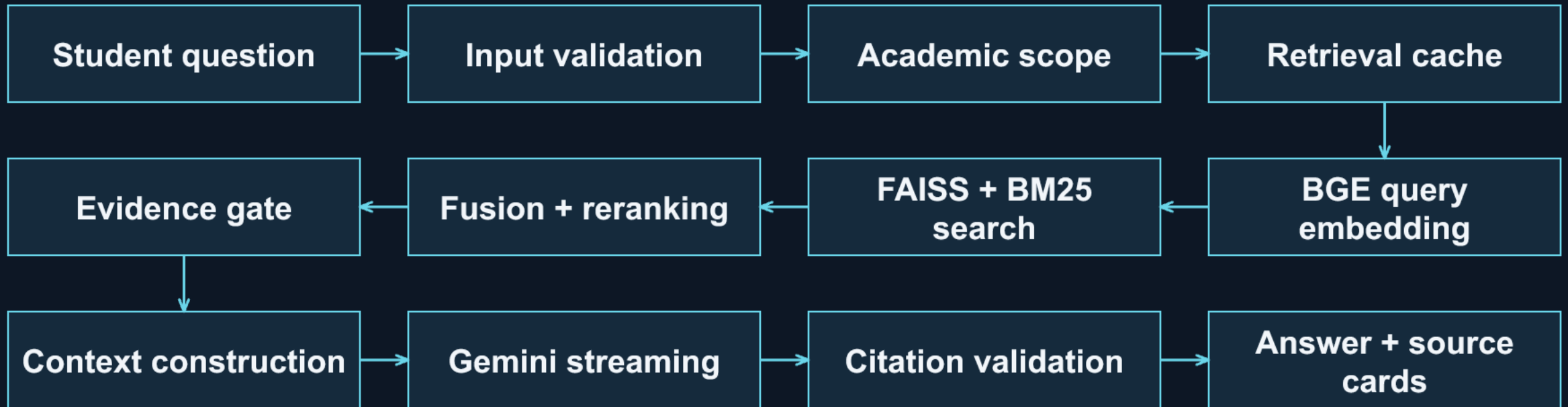


QUESTION-TIME PROCESSING



One Python application. No separate frontend/backend REST service.

Question processing workflow



SHORTER PATHS

Greeting

Deterministic reply

Exact syllabus overview

Indexed table, no Gemini

Weak evidence

Refusal, no Gemini

Token-aware optimization

Original index

3,227

chunks

952 oversized inputs

Exceeded the 512-token window

Active candidate

5,835

chunks

0 oversized inputs

448-token complete-input limit

17/17 checks before and after

Metadata and special tokens count toward the limit. Course rows and headings stay intact.



Technology stack

Application	Python 3.11 and Streamlit
Generation	Gemini REST streaming through Requests
Semantic search	BGE-small-en-v1.5, Sentence Transformers and FAISS CPU
Keyword search	rank-bm25
Ingestion	Requests, Beautiful Soup and PyMuPDF
Chunking and support	LangChain text splitters, python-dotenv and Pytest

Optional and inactive: Qwen with Transformers or llama.cpp

Security, reliability and testing

Implemented protections

- Server-side key and ignored .env
- PACE host and redirect validation
- PDF size and signature checks
- Prompt-injection instruction
- Citation-label validation
- Checksums and gated activation
- Non-destructive index rollback
- Localhost-only binding

89

automated tests passed

17/17 curated retrieval checks

- Missing R99 and unrelated queries rejected
- AIML R23 retained PDF pages 2 and 3
- Live Gemini placement answer validated

Observed results, not a guarantee of perfect accuracy or fixed latency.

Demonstration and next steps

II Year - I Semester [S2] Database Management Systems; Database Management Systems Lab; Digital Logic & Computer Organization; Discrete Mathematics & Graph Theory; Environmental Science; Object Oriented Programming Through Java; Python Programming.

II Year - II Semester [S2] Probability & Statistics; Artificial Intelligence; Operating Systems; Machine Learning; Design Thinking & Innovation; Operating Systems Lab; Full Stack Development-I.

This is a partial overview of the cited pages. Open the official PDF for all semesters, credits, electives, and detailed units.

2 official sources · 0.0s



Live demonstration

Open PACE AI
Ask “AIML R23 syllabus”
Show semester overview
Open PDF pages 2 and 3
Ask for missing R99
Show the safe refusal

Current limitations

- Saved snapshot, no automatic refresh
- Cold first semantic query, no OCR
- No auth or persistent chat history
- Small curated evaluation set

Future scope (not implemented)

- Conditional refresh and embedding reuse
- Gated updates, warmup and OCR
- Bounded history, auth and rate limits
- Larger evaluation set



PACE AI

A searchable, source-linked assistant for official campus information

Academic scope and document identity remain central.
Hybrid retrieval, token-safe embeddings and Gemini support the answer.
Citations and evaluated index selection make the evidence inspectable.

Thank You

Questions?